FF-S-325(COM-NBS)
Interim Amendment-1
March 23, 1961

FEDERAL SPECIFICATION

SHIELD, EXPANSION; NAIL, EXPANSION; AND NAIL, DRIVE SCREW

(DEVICES, ANCHORING, MASONRY)

This Interim Amendment was developed by the National Bureau of Standards, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity.

Para. 1.2.1: At the end of GROUP II, add the following:

"Type 4 - Expansion double-wedge anchors".

"Class 1 - (fig. 28)"
"Class 2 - (fig. 29)"

Para. 1.3: Revise by adding the following before "as specified":

"and XVIII,".

After para. 3.2.2.3.5, page 11, add the following:

"3.2.2.4 Type 4, expansion double-wedge anchors. - Unless otherwise specified, the material shall be at the option of the manufacturer. The anchors shall withstand the pulling proof test loads as specified in table XVIII, when set in concrete (see 4.3.2), in accordance with the recommendations of the manufacturer."

"3.2.2.4.1 Class 1 - This type of expansion anchor is shown in figure 28. As the nut is tightened against the washer, the shank is pulled upward. By means of internal prongs on the washer, the wedges are kept from moving upward with the shank. The taper on the shank thus forces the wedges against the sides of the hole, thus anchoring the assembly in position. On the expansion anchor shown, a standard hexagon nut may be used.".

"3.2.2.4.2 Class 2. - This type of expansion anchor is shown in figure 29. As shown in the figure, the two wedges and the connecting loop are one piece. One end of the spring is against the loop while the other end is in a drilled hole in the shank. Compression of the spring forces the wedges toward the unthreaded end of the shank. When inserted in a hole drilled in accordance

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with the recommendations of the manufacturer, pulled back slightly, and released, the shank will be anchored in the hole."

Para. 4.3.2: Add the following at end of first sentence in paragraph:

"and XVIII."

Para. 6.4: Revise "3.2.2.2.3.2", to read "3.2.2.3.2", and following this insert "3.2.2.4".

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TABLE XVIII - Group II, type 4 anchor

Class	Shank diameter	Nominal hole dimensions		Proof test loads a/
		diameter	depth	
	in.	in.	in.	pounds
1 (fig. 28)	1/4	1/4	1 7/8	1200
	5/16	5/16	2	1800
	3/8	3/8	2	1950
	1/2	1/2	3 1/2	3500
	5/8	5/8	3 1/2	4500
	3/4	3/4	4	5100
	7/8	7/8	4	6000
	1	1	4	7000
2 (fig. 29)	1/4	1/4	1 5/8	1200
	1/4	1/4	2 3/4	1600
	3/8	3/8	2 1/2	2000

a/ Recommended safe working load is one-fourth proof test load.

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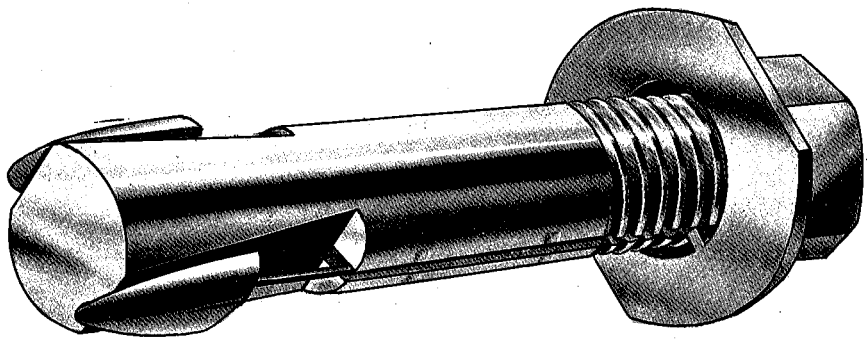


FIGURE 28. - Expansion double-wedge anchor, class 1.



FIGURE 29. - Expansion double-wedge anchor, class 2.

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